

NASHVILLE INTL, TN, USA

WMO: 723270

Lat: 36.119N

Lon: 86.689W

Elev: 183

StdP: 99.15

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13897

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.5	(c) -7.0	(d) -17.1	(e) 0.9	(f) -5.9	(g) -14.5	(h) 1.1	(i) -5.2	(j) 10.7	(k) 11.6	(l) 9.2	(m) 11.8	(n) 3.0	(o) 330	(p) 0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	34.7	23.7	33.5	23.6	32.4	23.3	25.6	31.6	25.0	30.8	24.5	29.9	3.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.3	28.2	23.3	18.5	27.7	22.8	17.9	27.3	79.4	31.9	76.8	30.8	74.6	30.1	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.6	(b) 7.6	(c) 6.8	(e) -12.7	(f) 36.5	(g) 3.0	(h) 1.9	(i) -14.9	(j) 37.9	(k) -16.6	(l) 39.0	(m) -18.3	(n) 40.1	(o) -20.5	(p) 41.5		
(a) 8.6	(b) 7.6	(c) 6.8	(e) -13.6	(f) 26.6	(g) 2.8	(h) 0.9	(i) -15.6	(j) 27.2	(k) -17.3	(l) 27.8	(m) -18.8	(n) 28.3	(o) -20.9	(p) 28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	3.8	6.1	10.5	15.8	20.5	24.8	26.6	26.3	22.7	16.4	10.1	5.8
	DBStd	9.27	6.50	6.05	5.56	4.76	3.93	2.68	2.14	2.32	3.55	4.58	5.03	5.65
	HDD10.0	638	211	137	62	9	0	0	0	0	0	5	61	154
	HDD18.3	1906	450	343	247	102	26	1	0	0	7	92	251	388
	CDD10.0	2774	19	28	79	184	326	444	515	506	381	203	63	26
	CDD18.3	1000	0	1	5	27	93	196	257	247	139	32	3	1
	CDH23.3	9679	0	4	47	243	821	1895	2647	2486	1238	279	18	1
	CDH26.7	3897	0	0	4	43	254	782	1167	1102	478	66	2	0
Wind	WSAvg	3.0	3.5	3.6	3.7	3.5	2.9	2.5	2.4	2.3	2.4	2.7	3.1	3.3
Precipitation	PrecAvg	1294	127	109	126	113	126	100	112	90	89	82	91	129
	PrecMax	1545	249	253	268	201	397	271	212	217	221	209	157	313
	PrecMin	978	54	52	70	22	13	2	52	35	5	0	15	60
	PrecStd	145	41	47	47	51	72	55	44	45	52	48	39	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.8	23.5	27.1	29.7	32.6	34.8	36.2	36.9	34.9	30.9	25.9	22.1
		MCWB	15.9	16.6	17.5	19.1	22.2	23.2	24.6	22.8	22.4	21.2	17.9	16.6
	2%	DB	18.6	20.6	24.6	28.0	30.9	33.4	34.3	34.8	32.9	28.4	22.7	19.4
		MCWB	14.9	14.7	16.0	18.1	21.1	23.5	24.4	23.7	22.1	19.8	16.0	15.6
	5%	DB	16.6	18.2	22.3	26.2	29.5	32.2	33.2	33.3	31.4	26.4	20.7	17.0
		MCWB	13.6	13.2	14.7	17.4	20.5	23.1	24.2	23.6	21.8	18.4	15.1	13.8
	10%	DB	14.2	15.9	20.0	24.4	28.0	31.0	32.1	31.9	29.6	24.4	18.6	15.0
		MCWB	10.8	11.4	13.3	16.7	20.2	22.4	23.7	23.2	21.1	17.8	14.0	12.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.6	18.0	18.8	21.3	24.0	25.7	26.5	26.8	24.5	22.9	19.4	18.3
		MCDB	19.4	21.7	24.5	27.1	29.0	31.7	32.4	32.3	30.4	27.8	22.9	21.0
	2%	WB	15.7	16.5	17.4	20.1	22.8	24.9	25.7	25.6	23.7	21.6	17.9	16.4
		MCDB	17.9	19.9	22.6	25.3	28.4	31.0	31.8	31.2	29.2	25.9	21.5	18.8
	5%	WB	14.1	14.4	16.1	18.9	21.9	24.2	25.2	24.8	23.0	20.4	16.4	14.5
		MCDB	16.8	17.0	20.7	23.8	27.4	29.9	31.2	30.2	28.1	24.1	19.7	16.4
	10%	WB	11.4	12.2	14.7	17.8	21.1	23.6	24.6	24.2	22.3	19.2	14.8	12.2
		MCDB	13.6	15.1	19.0	22.6	26.2	28.9	30.2	29.3	27.2	23.1	17.7	14.7
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.2	11.3	11.8	10.9	10.5	10.0	10.3	11.2	11.7	11.0	9.2
		MCDBR	11.3	12.9	14.2	13.7	12.7	11.9	11.2	12.0	12.9	13.8	13.5	11.2
		MCWBR	8.3	8.7	7.8	6.3	5.1	4.1	3.6	3.8	4.3	6.4	8.0	8.3
	5% WB	MCDBR	10.0	10.7	12.1	11.0	10.6	10.4	10.1	10.0	10.3	10.4	10.4	9.7
		MCWBR	8.2	8.6	7.8	6.1	5.1	4.3	3.8	3.8	4.2	5.7	7.6	8.2
Clear-Sky Solar Irradiance	taub	0.319	0.326	0.373	0.402	0.452	0.477	0.509	0.521	0.430	0.374	0.335	0.311	
	taud	2.478	2.449	2.347	2.265	2.172	2.138	2.067	2.021	2.261	2.404	2.467	2.514	
	Ebn at Noon	878	914	893	881	839	815	785	766	823	846	850	865	
	Edh at Noon	84	96	117	134	149	154	164	169	126	100	84	76	
All-Sky Solar Radiation	RadAvg	2.21	2.78	3.89	5.10	5.71	6.25	5.95	5.49	4.68	3.63	2.61	1.90	
	RadStd	0.19	0.35	0.36	0.35	0.47	0.42	0.32	0.26	0.39	0.45	0.27	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+151	+101
(47) Regional (19 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+98

Nomenclature: See separate page